T. A. MARRYSHOW COMMUNITY COLLEGE

DEPARTMENT OF INFORMATION TECHNOLOGY

LAB NAME: LAB 11 CREATING A WINDOWS VIRTUAL MACHINE

COURSE NAME AND NUMBER: DATA SECURITY CONCEPTS CIT245

DATE: MARCH 2026

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LAB-OBJECTIVE

The goal was to create a Windows 11 virtual machine using VirtualBox. This involved downloading the Windows 11 ISO from Microsoft, configuring the VM settings (name, RAM, storage), and launching the VM to verify it boots correctly.

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PRE-LAB QUESTIONS (VIRTUALIZATION BASICS)

1. **Define the following:**
    - **Hypervisor:** Software that creates and runs virtual machines by managing physical hardware resources.
    - **Virtual machine (VM):** A software-based computer that runs its own OS and applications using virtualized hardware.
    - **Virtual hard disk (VHD/VMDK):** A file that acts as a hard drive for a VM, storing the OS and data.
    - **Guest operating system:** The OS running inside a VM (e.g., Windows 11).

2. **Why do IT professionals commonly use virtual machines for labs, testing, and security training? Provide at least two reasons.**
    - **Isolation:** VMs are sandboxed, so malware or system failures in a VM do not affect the host computer.
    - **Efficiency:** One physical machine can run multiple VMs, saving hardware costs and allowing quick setup of test environments.

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CREATING A WINDOWS VIRTUAL MACHINE

1. **Record the virtual machine software you used:**
   Oracle VirtualBox

2. **Record the main VM configuration settings you selected:**
   - **VM Name:** Win11VM
   - **RAM assigned:** 4 GB (4096 MB)
   - **CPU cores:** 2
   - **Disk size:** 80 GB
   - **Network mode:** NAT
   - **ISO file used:** Windows 11 (multi-edition ISO for x64 devices)

3. **Describe what happened when you started the VM and booted from the Windows installer ISO.**
   The VM window opened and booted from the ISO. The Windows logo appeared, followed by the language selection screen, confirming the installation media loaded successfully.

4. **What challenges did you encounter while creating the VM (if any)? How did you solve them?**
   No major challenges. The ISO download took time, but I let it run in the background. The setup process in VirtualBox was straightforward.

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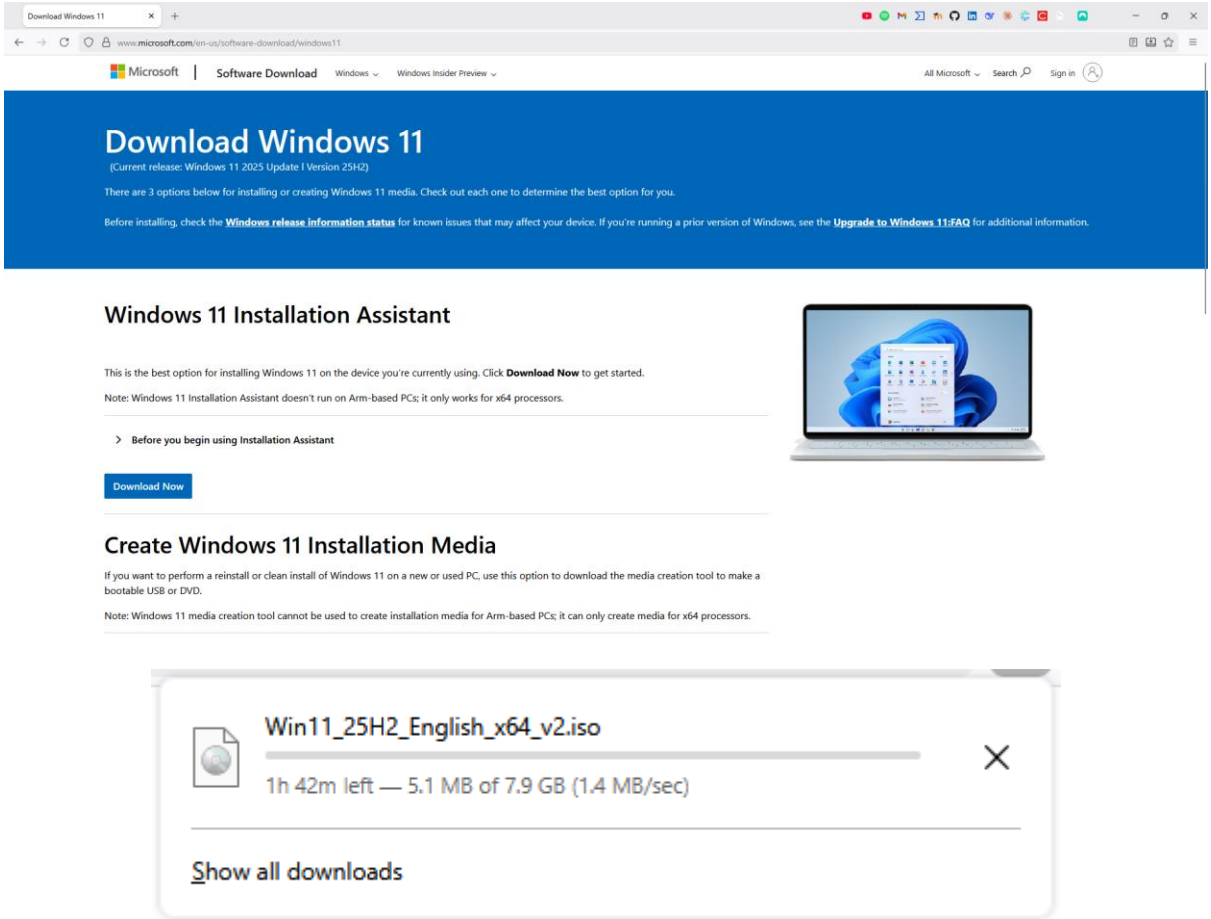
EVIDENCE & SCREENSHOTS

Windows 11 ISO Download



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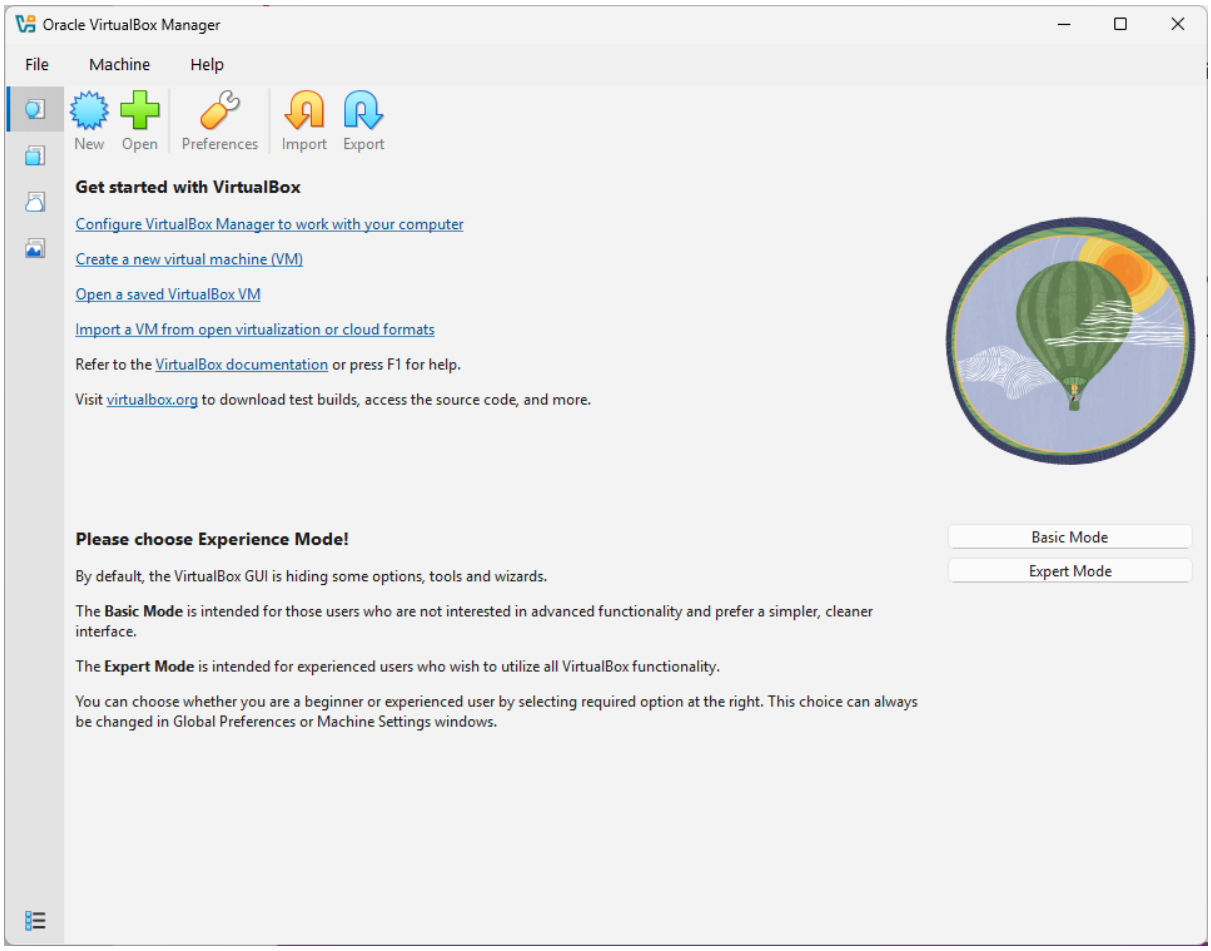
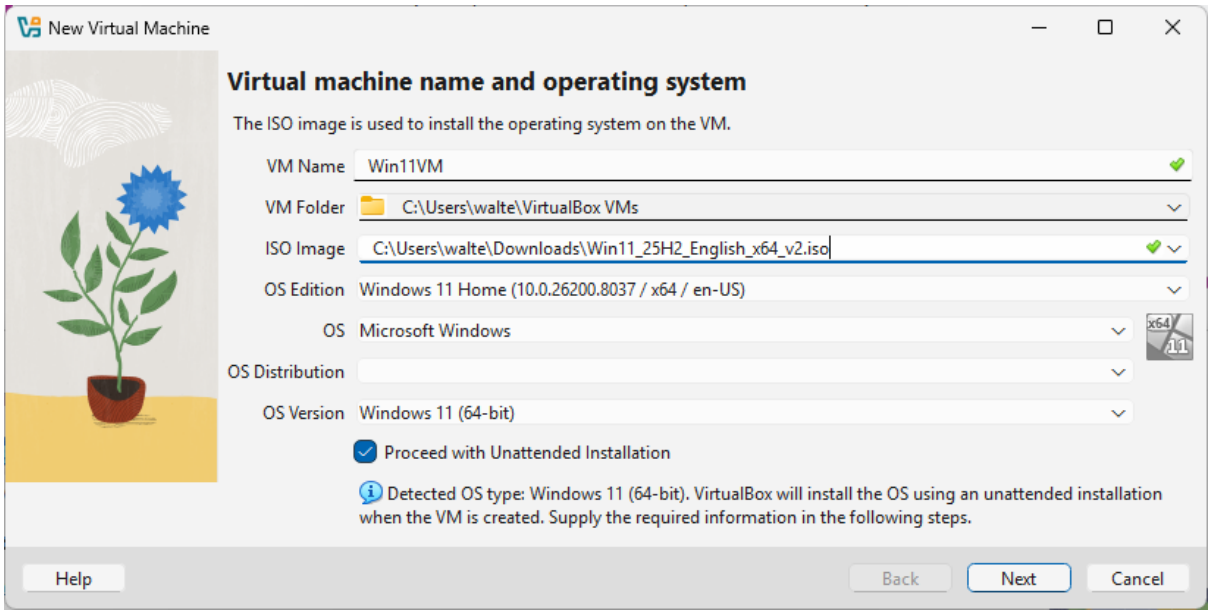
VirtualBox - VM Name and ISO Selection





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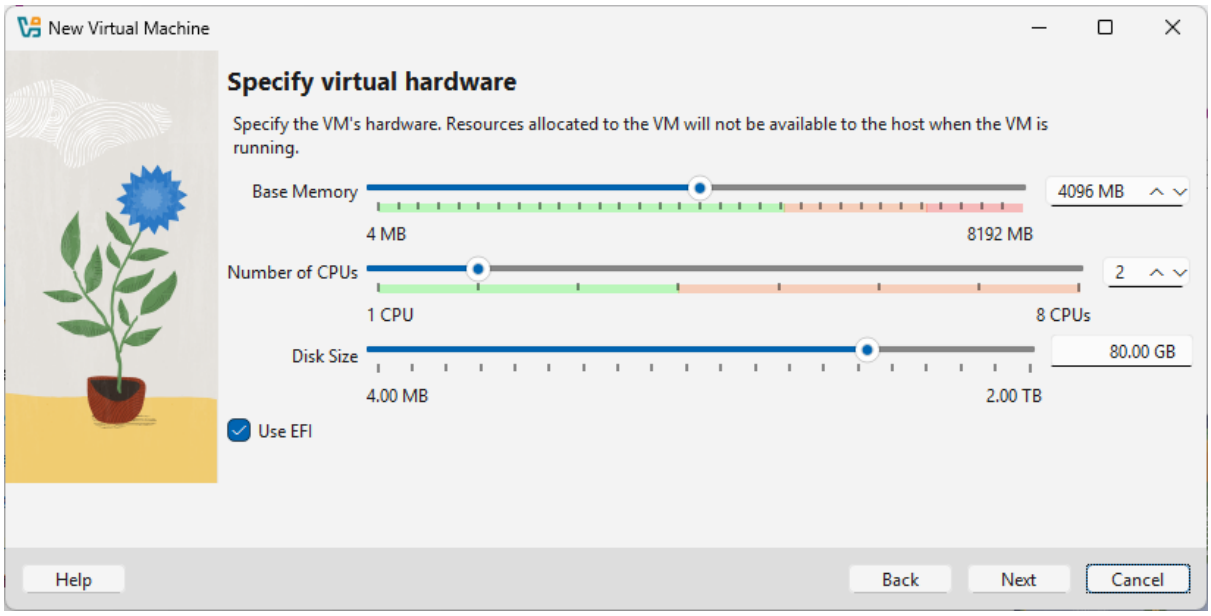
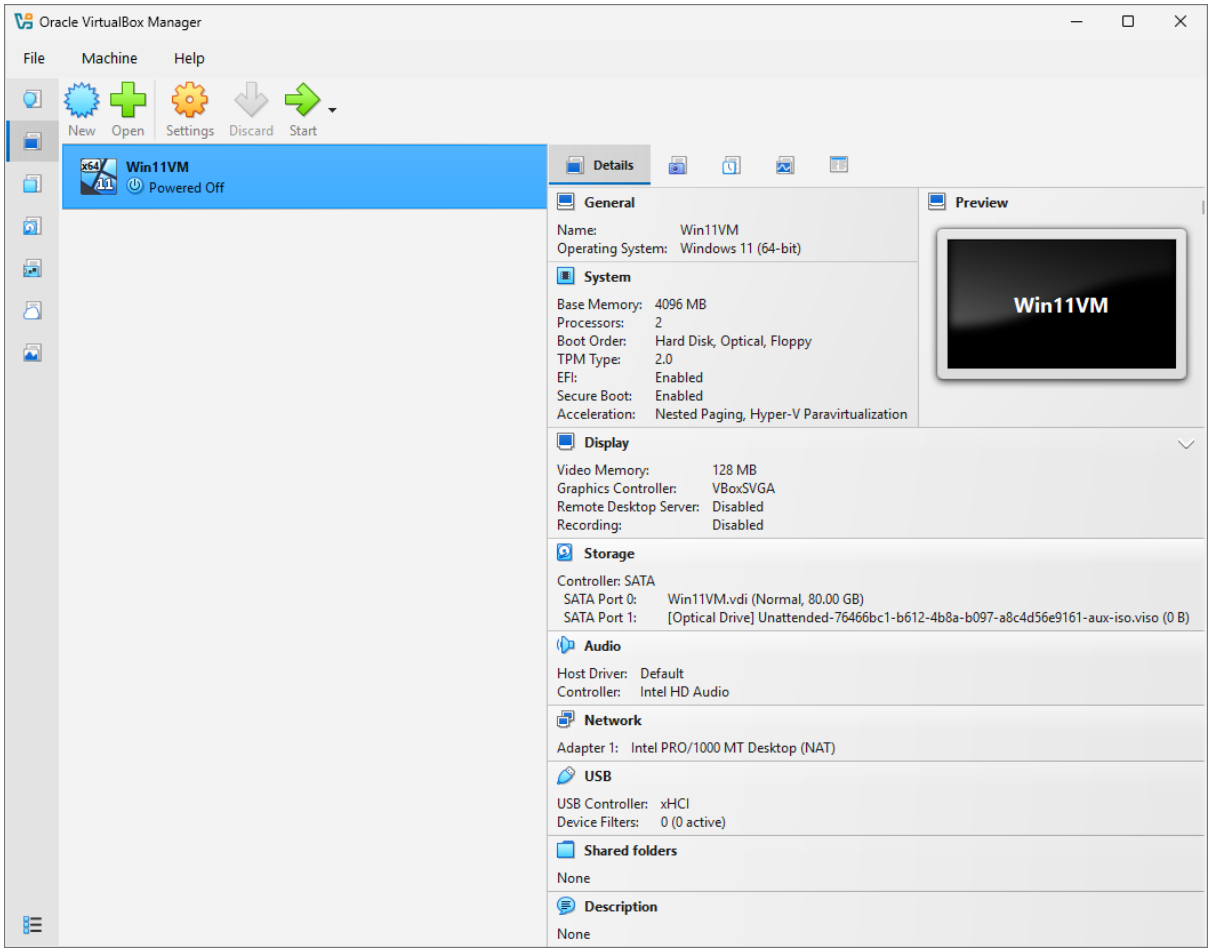
VirtualBox – Hardware & Virtual Hard Disk Configuration



VirtualBox Main Window with Win11VM



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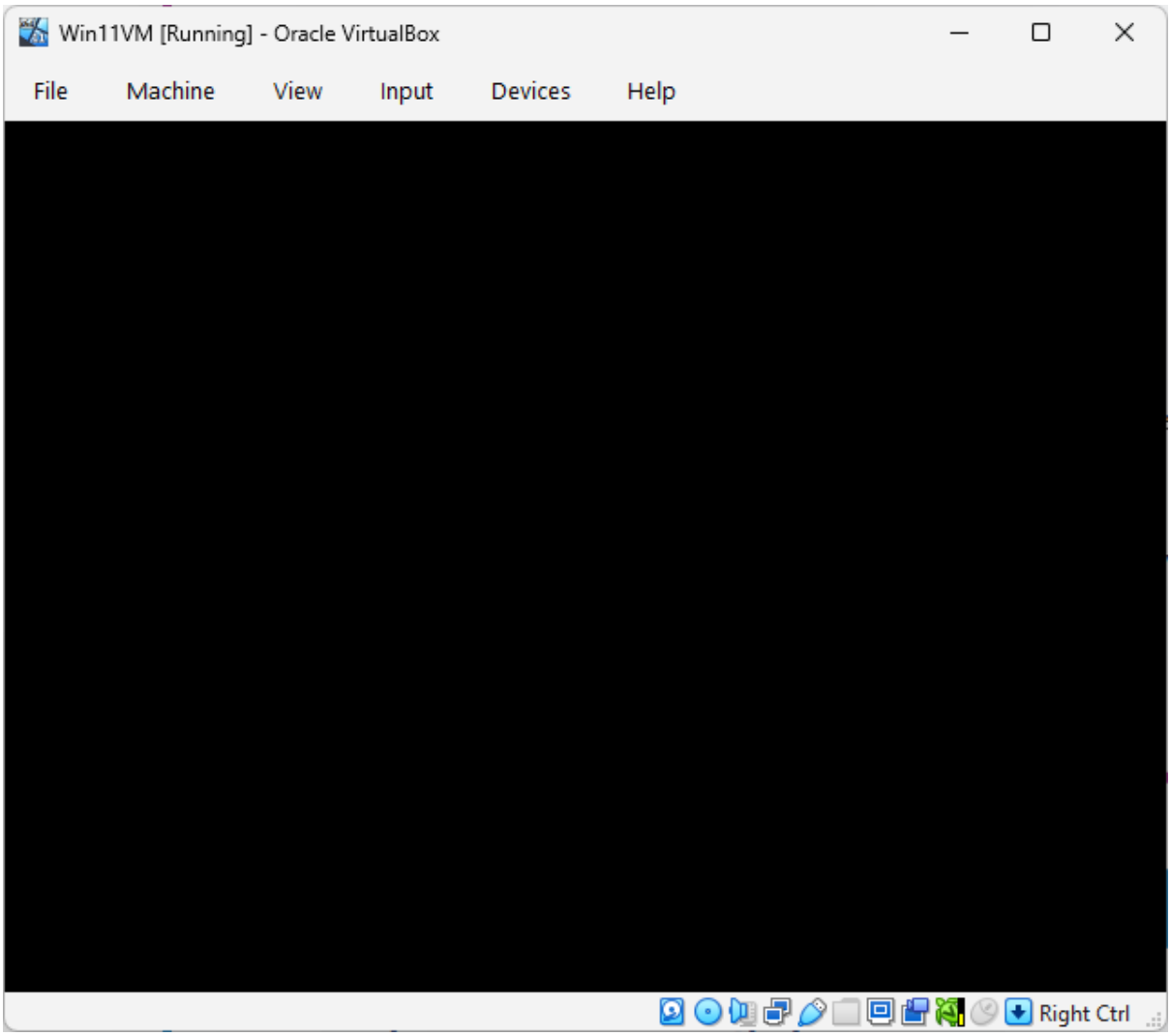
VM Booting



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OVERALL ANALYSIS & REFLECTION

1. **Summarise why virtualization is an important tool in cybersecurity and IT operations.**

   Virtualization allows safe testing in isolated environments, saves hardware costs, and enables quick deployment of systems for training, development, and security analysis without risking production networks.

2. **What did you learn while completing this project? Mention at least two insights.**
   - I learned how to download the correct Windows ISO from Microsoft and attach it to a VM.
   - I learned that a VM requires proper hardware settings (RAM, CPU, storage) to run efficiently, and the boot order must be set to the optical drive to start the installation.

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